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## 2019-Ph-H1-Q5

### Section: Electricity

### Topic: Current, PD, Power, Resistance

#### Question:

Two resistors of resistance  $R$  are connected in parallel. What is the equivalent resistance?

- A)  $R$
- B)  $2R$
- C)  $R/2$
- D)  $R/4$
- E)  $\sqrt{R}$

#### Answer:

For parallel resistors:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R} + \frac{1}{R} = \frac{2}{R} \Rightarrow R_{\text{eq}} = \frac{R}{2}$$

#### Correct Option: C

#### Guidance for Students:

Resistors in parallel **decrease** the total resistance — the current has more than one path.

#### Revision Tips:

- Parallel:  $\frac{1}{R_{\text{total}}} = \sum \frac{1}{R_i}$
  - Two equal resistors in parallel = half the resistance
  - Series: resistances **add**
  - Parallel: total resistance is always **less than the smallest resistor**
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